Research Proposal

Title

Impact of Uber's Acceptance and Cancellation Rate Algorithms on Driver and Passenger Outcomes.

1. Introduction and Background

The current rideshare industry in Australia, in general, has changed a lot by replacing the traditional way of booking a ride with AI or the digitalization of the platforms used to offer services, which is changing the ways of delivering and managing the services offered on them. Ride-sharing services such as Uber use algorithms to coordinate drivers and passengers, assign them trips, set their fares, evaluate the performance of drivers, and control different processes of the service delivery. This way of managing is called algorithmic management.

The influence of Uber's algorithmic management system on drivers could be explained by using such mechanisms as assignment of trips, acceptance rate, cancellation rate, driver ratings, customer ratings, and others.

Acceptance and cancellation rates could be especially important, as they reflect a measurable part of the interaction between drivers and the platform. The decision about accepting or rejecting a booking could depend on several things, including trip length, estimated fare, location, destination, waiting period, and earnings. Cancellation could make the driver's and passenger's experience worse, as it could increase the waiting time and force passengers to ask for another car.

Thus, the research is devoted to the study of the algorithmic management system of Uber, and the key indicators of the research will be acceptance and cancellation rates. The study will include quantitative analysis of secondary data and qualitative investigation of the drivers' experience.

2. Research Problem

In the case of Uber, acceptance and cancellation requirements can cause Uber drivers to accept those trips they find inappropriate and not profitable enough. The distance of a trip, fare, location, waiting period, and profitability of a booking may determine whether a driver accepts or rejects it. Drivers can cancel booked trips depending on certain conditions or if a trip does not fit their expectations. These actions have consequences for customers. Drivers' cancellations can cause longer waiting periods, repeated bookings, and uncertainty about whether a customer will be served according to his/her requests. Meanwhile, drivers may face financial stress, loss of autonomy or worries about algorithms used in making their decisions.

Thus, it is necessary to conduct research on algorithmic management at Uber from both quantitative and qualitative angles. The aim of this project is to investigate quantitative relationships between the acceptance, cancellation, and booking results as well as drivers' opinions on Uber's algorithmic policy. Further, when there are repeated cancellations, it sometimes happens that the reversal of the transaction causes issues, as in some cases customers had paid for a booking and reversal takes time, which causes more effort for the bank and Uber.

3. Research Aim

To examine the effects of Uber's algorithmic management, especially through acceptance and decline rate algorithms, on the behavior of drivers as well as passengers' performance through a combination of quantitative data analysis and perception of drivers.

4. Research Objectives

The research will achieve the following objectives:

a. Investigate the relationship between acceptance rate and driver behaviour.

b. Examine cancellation patterns and their relationship with passenger booking outcomes.

c. Understand Uber drivers' perceptions of acceptance and cancellation policies.

d. Integrate quantitative and qualitative findings to develop a more comprehensive understanding of Uber's algorithmic management.

5. Research Questions

5.1 Main Research Question

- **What is the impact of Uber's acceptance and cancellation rate algorithms on driver and passenger outcomes?**

5.2 Supporting Research Questions

- What factors are associated with acceptance rates and driver behaviour?
- What factors are associated with driver cancellations and passenger booking outcomes?
- How do Uber drivers perceive acceptance and cancellation requirements?
- What effects do drivers perceive algorithmic management to have on fairness, transparency and autonomy?

6. Literature Review

6.1 Algorithmic Management in the Gig Economy

Algorithmic management has emerged as an essential characteristic of the platform-based gig economy espicially in Uber platform. In the context of digital platforms, algorithms are used to assign tasks to workers, monitor their performance, assess the workers and control their behavior. In contrast to regular employment, algorithmic management means that workers may be provided with instructions and evaluations of their performance without interaction with a human manager.

Algorithms play a significant role in ride-sharing services, allocating orders to specific drivers, making assignments of trips and assessing various features of performance of drivers.

An example of Uber is essential because the platform matches a considerable number of drivers and customers using algorithms.

6.2 Impact on Driver Behavior

There are different reasons why drivers decide whether to accept a particular trip, among which there can be such aspects as the distance, payment, estimated travel time, destination point, and earnings from the trip.

In this regard, the behavior of drivers' acceptance of trips can be considered the result of the interaction of certain characteristics of the platform and the driver himself. In case drivers find some trips not profitable enough for them, they will be more likely to reject such offers.

6.3 Passenger Outcomes

The implications of algorithmic management extend beyond the effects on drivers. The experience of the passenger could also be impacted by the acceptance of the driver and the behaviour when it comes to cancelling the booking.

The long waiting time for the passenger could be due to the cancellation of the booking, thus affecting the efficiency of the platform and the level of passenger satisfaction. The study will also explore passenger-specific implications that could be found in the data set.

6.4 Human Experiences of Algorithm Management

Quantitative metrics alone are not sufficient to capture the experience of work under algorithmic management. The driver might have his/her own perception regarding the fairness, transparency, autonomy, economic stress, and satisfaction with the platform's decision.

Measurable and Human Outcomes

Measurable outcomes

- Acceptance rate
- Cancellation rate
- Trip fare
- Distance

- Earnings
- Waiting time
- Booking outcome
- Ratings

Human experiences

- Fairness
- Transparency
- Autonomy
- Financial pressure
- Satisfaction
- Perceptions of algorithmic decisions
-

7. Research Gap

The existing literature about platform work and algorithmic management offers good information about the relationship between algorithms, workers, and the efficiency of the platform. Yet, studies relying only on quantitative behavioral data do not always

fully account for workers' perceptions of the algorithmic systems that affect their decision-making process. On the other hand, qualitative interview-based studies offer a thorough picture of workers' experience; however, they do not prove any behavioral patterns in a large dataset. This study tries to bridge the gap between quantitative and qualitative approaches.

Academic Contribution

The study aims to:

- Extend understanding of algorithmic management within the Uber platform.
- Examine the relationship between acceptance, cancellation and booking outcomes.
- Connect behavioural data with driver perceptions.
- Address limitations associated with relying on a single research method.
- Provide a more comprehensive view of algorithmic management by combining measurable outcomes with human experiences.

Practical Contribution

The findings may help highlight the need for a balance between:

Platform Efficiency → Driver Welfare →Passenger Experience

8. Research Methodology

8.1 Research Design

This research will adopt a mixed-methods design. In terms of quantification, secondary data will be analysed for patterns and relationships between booking features and driver behavior on booking outcomes. As for the qualitative research, the study will explore how drivers perceive the acceptance and cancellation rules set by Uber using either an interview method or by asking open-ended questions of the respondents.

8.2 Quantitative Data

Dataset: ncr_ride_bookings.csv

Dataset size: Approximately 105,000 records

The dataset includes the data of ride-booking, which can be analyzed to identify the reasons that lead to booking and cancelling of rides.

8.3 Data Sampling

This research is based on the large secondary dataset from Kaggle, which consists of about 105,000 bookings.

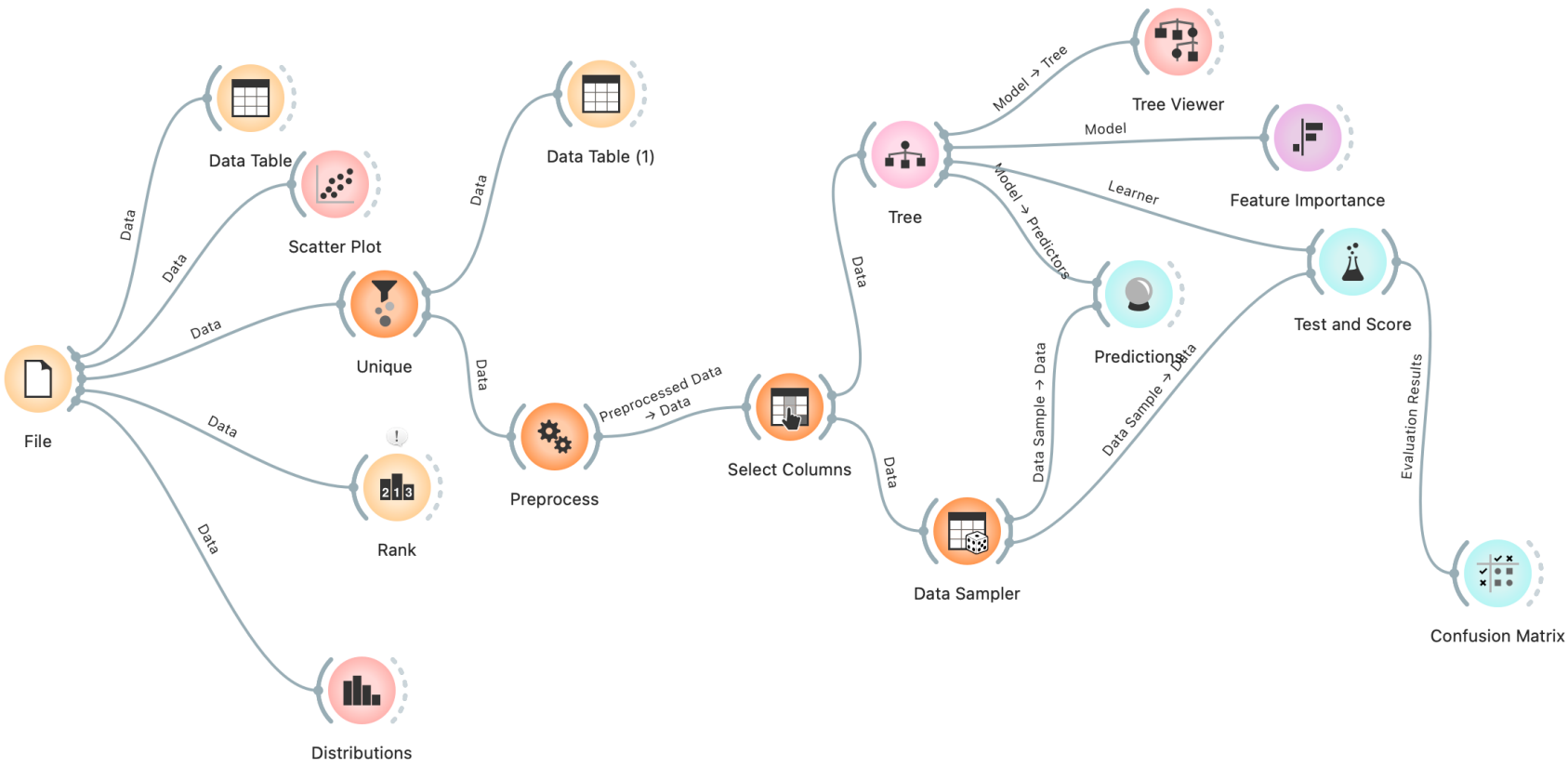
The results will be considered patterns in the dataset being studied and not universal evidence of all Uber users.

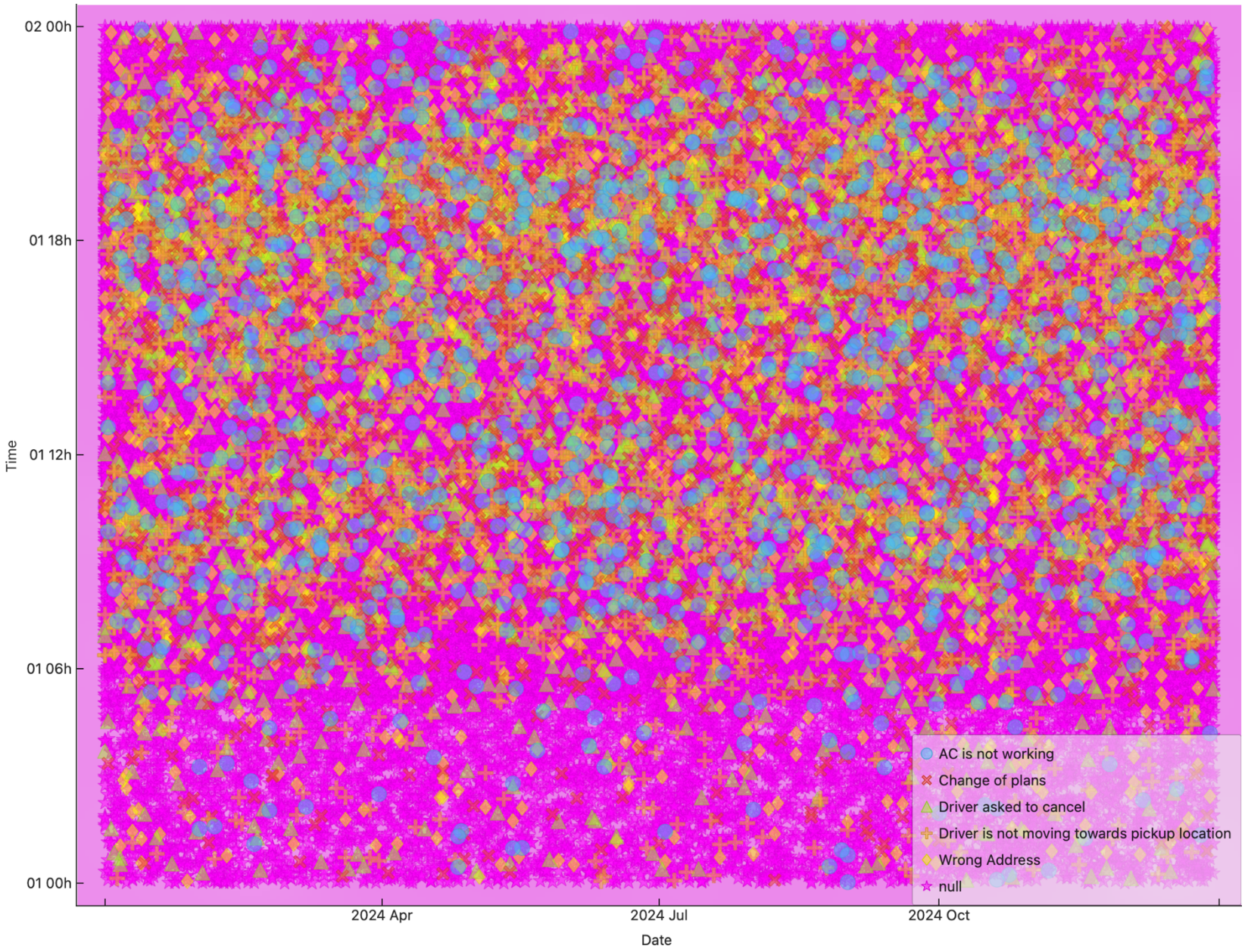
9. Data Preparation and Analysis

9.1 Orange Data Mining

Orange Data Mining, a free visual analytics and machine learning platform, will be used for preparing and analyzing the quantitative data. The proposed pipeline will comprise the following steps:

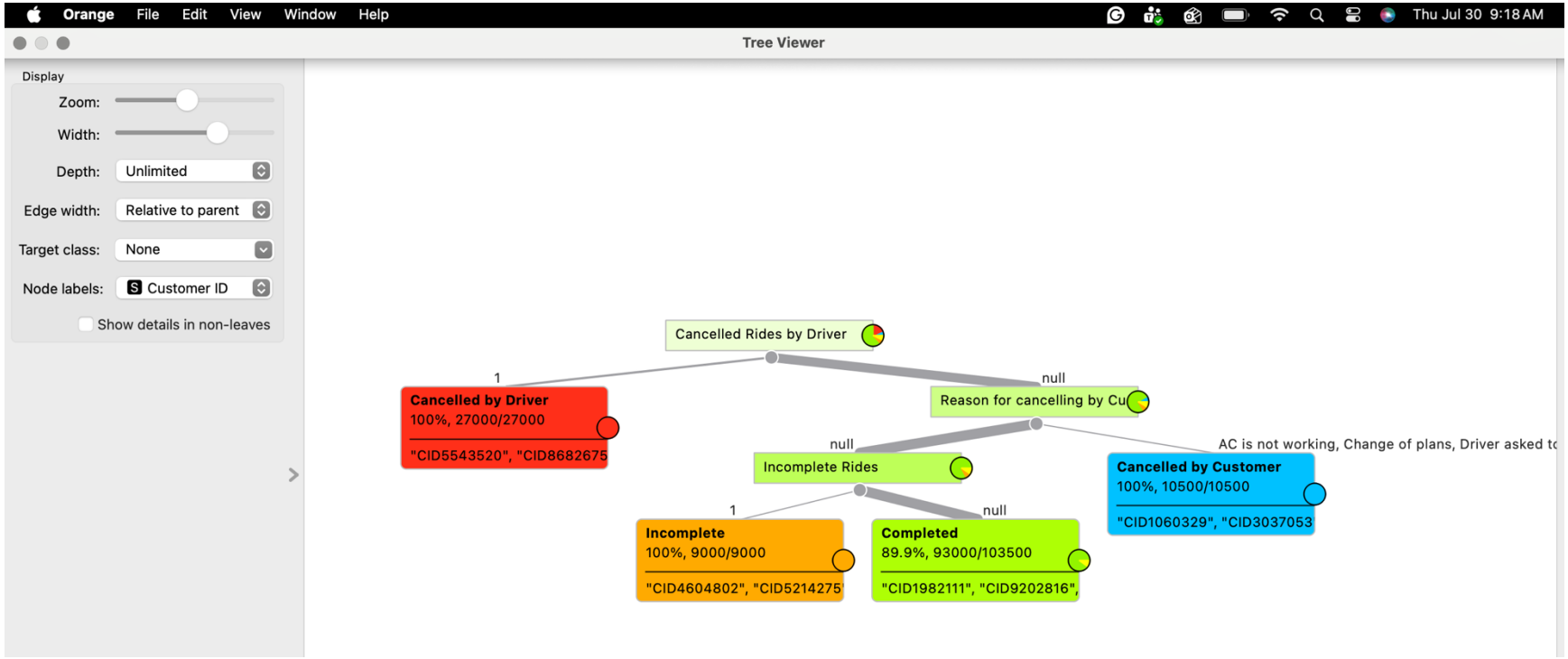
Data set -> Data preparation -> Feature selection/feature transformation -> Decision tree -> Predictions -> Confusion matrix -> Evaluation

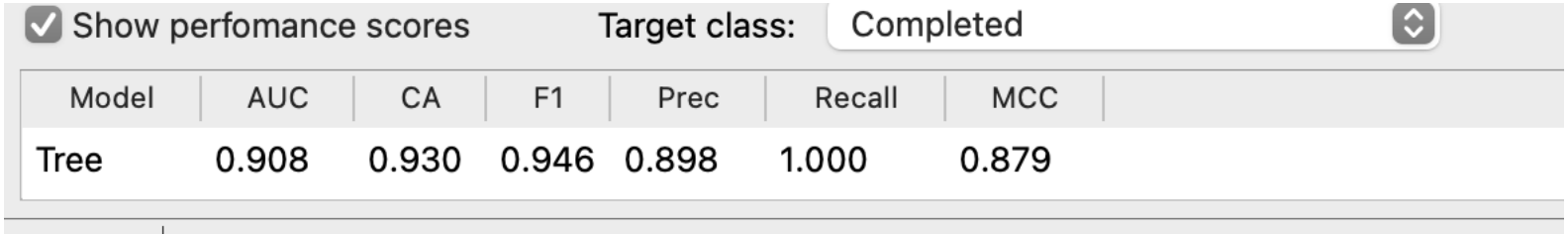

File
Data Table
Scatter Plot
Unique
Rank
Distributions
Data Table (1)
Preprocess
Preprocessed Data → Data
Select Columns
Tree
Tree Viewer
Model → Tree
Model
Feature Importance
Learner
Model → Predictors
Predictions
Test and Score
Data Sampler
Data Sample → Data
Data Sample → Data
Evaluation Results
Confusion Matrix
Data


02 00h
01 18h
01 12h
01 06h
01 00h
Time
2024 Apr
2024 Jul
2024 Oct
Date
AC is not working
Change of plans
Driver asked to cancel
Driver is not moving towards pickup location
Wrong Address
null

9.2 Decision Tree Analysis

Decision Tree analysis will be conducted to reveal relationships between selected predictor variables and target outcome. This may reveal certain combinations of variables related to specific booking results. At the same time, the Decision Tree will be used as a tool for prediction and exploration, but it will not be viewed as proof of a direct causal relationship between particular variables.





9.3 Confusion Matrix and Preliminary Results

The Confusion Matrix will be utilized for the purpose of assessing the classification results of the decision tree model.

Based on the preliminary assessment, 32,587 instances were classified as being unable to reach the target outcome.

The above result cannot be viewed as the failure of Uber's algorithm in 32,587 instances since it is the number of instances from the analyzed dataset that were classified as being unable to achieve the target outcome.

Confusion Matrix

Actual \ Predicted	Cancelled by Customer	Cancelled by Driver	Completed	Incomplete	No Driver Found	Σ
Cancelled by Customer	7303	0	0	0	0	**7303**
Cancelled by Driver	0	18954	0	0	0	**18954**
Completed	0	0	65016	0	0	**65016**
Incomplete	0	0	0	6330	0	**6330**
No Driver Found	0	0	7397	0	0	**7397**
Σ	**7303**	**18954**	**72413**	**6330**	**0**	**105000**

10. Qualitative Data Collection

The qualitative element will include participation of Uber drivers via semi-structured interviews or open questions.

- Acceptance requirements
- Cancellation requirements
- Trip allocation
- Earnings and financial pressure
- Fairness
- Transparency

- Driver autonomy
- Perceived consequences of rejecting or cancelling trips
- Overall satisfaction with algorithmic management

Qualitative data will be analyzed thematically in order to determine common themes among driver experiences.

Then, the qualitative results will be compared with the quantitative results.

11. Integration of Quantitative and Qualitative Findings

The last phase in the mixed-methods research involves combining the two types of evidence.

In the quantitative evidence phase, answers to questions like the following will be sought:

What patterns can be detected?

On the other hand, answers to questions such as the following will be provided by the qualitative evidence phase:

Why is this pattern being exhibited, and what is the driver's experience on that matter?

For instance:

Quantitative evidence:

Certain features of trips are linked to higher cancellation rates.

Qualitative evidence:

Drivers would explain why certain trip features are a problem for earning, distance, or time.

Thus, combining both pieces of evidence will allow for a better comprehension of algorithmic management at Uber.

12. Validity, Reliability and Limitations

The validity of this research is proven by means of utilizing the dataset that contains more than 105,000 records and the process of conducting a data analysis via Orange Data Mining. As such, the Decision Tree presents a picture of relationships between variables and the target, while the Confusion Matrix presents a way to assess the predictive ability of the trained model. At the same time, there is a limitation of the dataset as a quantitative approach to the research since it may show certain things that occurred in the data, but it cannot give a thorough answer on why certain actions were taken by the driver. Another limitation is related to the fact that the dataset is secondary data that was obtained from Kaggle. As such, the researcher cannot control the collection, sampling, geographical scope, and data quality. Therefore, the results of the research cannot be automatically generalized to all Uber drivers and passengers..

13. Expected Outcomes

The research is expected to give us a better insight into the connection between Uber's algorithmic management, driver behavior, and passenger bookings. The quantitative part of the research is expected to help identify patterns connected with acceptance rates, cancellations, and booking outcomes. The Decision Tree and the Confusion Matrix are expected to confirm whether or not we were able to select suitable predictors for our dependent variable. The qualitative part of the research is expected to bring up new questions on the experience of drivers under algorithmic management. Thus, the expected result of the research is not just an answer to the question but rather some new questions for further investigation. Finally, the purpose of this research is to help in creating more efficient and satisfying algorithms and driver-passenger experiences in the gig economy.

14. Research Significance

Importance of the Research:

This research is important due to its combination of data analysis and consideration of human experience in algorithmic management. The results of the quantitative analysis include the construction of a Decision Tree with predictions and a Confusion Matrix. These two methods give us an opportunity to analyze the relationship between certain variables and the target variable.

15. Proposed Timeline

12-WEEK GANTT CHART

Task	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12
Topic Selection and approval		■										
Literature review			■									
Research Question & Objectives				■								
Methodology design					■							
Survey/interview design						■						
Ethics Approval							■					
Data collection								■				
Data Analysis									■			
Findings and discussion writing										■		
Final report writing											■	
Submission												■
Presentation												■

16. Conclusion

Algorithmic management is to become the focus of the research to investigate how Uber manages its driver workforce by applying the concepts of acceptance and cancellation rate mechanisms.

As a result of preliminary quantitative analysis, about 32,587 out of 105,000 bookings did not reach the target outcome and represented about 31% of the total dataset used for Decision Tree model creation in Orange Data Mining.

However, this statistic must not be interpreted as 31% of all trips conducted by Uber are somehow failed by the algorithm. Instead, it can be concluded that 31% of the observations of the dataset used in the research did not reach the target outcome according to the Decision Tree model. Nevertheless, this finding represents a valuable starting point for further investigation.

In addition, behind each data point there can be found either a driver, waiting for a proper ride, or a passenger, waiting for his or her driver. In other words, it is necessary to consider the importance of the impact of algorithmic decisions on people's lives and their time, incomes and journeys. That is why it is not enough just to look at the algorithm's ability to predict the future events and its precision. Though, there is no doubt that the results of the Decision Tree and Confusion Matrix represent the valuable quantitative evidence. Nevertheless, they do not answer the question about the reasons why drivers react to algorithmic decisions in the way they do and what feelings and emotions are involved into the process of making decisions and experiencing different problems related to fairness, transparency, autonomy and financial pressure.

The suggested research approach can help to combine quantitative evidence with qualitative perspective of the problem under discussion

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